

MIE366 – DESIGN ASSIGNMENT ROADMAP

Department of Mechanical and Industrial Engineering – University of Toronto

The intent of the design assignments will be to mirror the actual development, building, and testing of a solution to a real product design problem which (after things return to normal) you could potentially build on your own. The design assignments will take you from initial design of a circuit, through a prototype phase, and into revision of the design as the last deliverables. Since these assignments build on one another, it is crucial that you be aware of the timeline and deliverables. Please read through this roadmap before starting DA1, and refer to it often throughout the course. Additional information and supplementary resources will be added to the assignments section to assist you with each step.

Design Assignment Timeline

The timeline posted below is **tentative** and may be updated to work around deliverables in other courses if we find that there is a period where most of the class is exceptionally busy.

First day of classes	Assignment 1 Posted <i>'Design Proposal'</i>	In this assignment, you will propose a circuit design that will form the <i>current sense stage</i> of your power supply. The deliverable will include: 1. Research into existing solutions and design patterns related to the problem. 2. Two proposed 'candidate' designs, based on the specification provided. 3. One final design, consisting of a schematic and a SPICE simulations demonstrating that all functionality is implemented (in theory)
September 21	Assignment 1 Due	TAs will begin work to provide individual feedback on your design as quickly as possible. Feedback will be delivered with sufficient time to incorporate changes into DA2.
	Assignment 2 Posted <i>'Prototype Design'</i>	In this assignment, you will produce a PCB layout for a prototype of the design you created in DA1. If we identify that your design from DA1 is fundamentally flawed in some way, a reference design will be available as a backup. The deliverable will most likely include: 1. A modified schematic incorporating any TA feedback / fixes from DA1 feedback, and a newly designed filter circuit. 2. A bill of materials with part numbers and related information. 3. A PCB design consisting of original CAD files and production-ready Gerber files.
October 4	Assignment 2 Due	We will begin to fabricate each of your custom PCBs shortly after submission for DA2 closes.
	Assignment 3 Posted <i>'Testing and Revision'</i>	In this assignment, you will build and test your prior PCB. We expect to offer a few scheduled build sessions where

you will solder your PCBs, as well as drop-in times for troubleshooting and testing. There will also be a window for you to produce an updated set of PCB files, enabling a “redo” of your PCB if you find serious design errors. Any redo version will be available to solder and use with DA4.

Deliverables include:

1. A set of test results from your first PCB and, if necessary, a revised set of design files (to be fabricated).
2. Results of testing and calibration of your current sense PCB.

November 15	Assignment 3 Due	Written deliverables due here, although please note that we may require updated Gerber files before this date if you intend to have your board remade. An update will be posted on Quercus in advance of any fabrication deadline.
	Assignment 4 Posted <i>‘Design Verification and Final Report’</i>	In the final assignment, you will integrate your design with a provided motherboard. This will include: 1. Attaching your completed design from DA3 to the provided motherboard and testing the result. 2. Producing an updated schematic, incorporating all changes and fixes to date. 3. Producing a final bill of materials as one would supply for PCB Assembly (“PCBA”). 4. (Optional) Creating a case design, which we will 3-D print for free, to house your project.
Last day of classes	Assignment 4 Due	We intend to provide post-course support for anyone who wishes to order personal designs from a PCB vendor. This would include access to and support in using our SMD PCB fabrication space (pick-and-place and oven) to build your designs.